

USV Registration GUI — Temp7.3 Release Note

Version: Temp7.3

Release date: August 2026

Overview

Temp7.3 improves the usability, portability, and internal organization of the USV Registration GUI.

The major updates include:

- Current working filename displayed in the window title
- Unsaved-change indicator ()
- Improved file loading and saving workflow
- Repository-relative dependency loading
- Reorganized GUI state management
- Updated JPEG export pipeline

1. Current File and Save Status

The GUI window now displays the name of the currently opened file.

Example:

USV260801_001.h5

When the annotation has been modified but not saved, an asterisk is appended:

USV260801_001.h5 *

The  appears when:

- A new USV box is added
- An existing box is moved
- An existing box is resized
- A box is deleted

After saving, the indicator is removed.

This makes it easier to identify the current recording and prevents accidental loss of unsaved annotations.

2. Improved File Loading

Files can be changed directly through the **Load** button without restarting the GUI.

When a new file is loaded:

- Spectrogram data are reloaded
- Existing USV annotations are loaded when opening a `.jld` file
- Current USV index is reset
- Time controls are recalculated for the new recording
- Histograms and spectrogram displays are refreshed
- Save status is reset

Supported input workflow:

```
.h5
|
├─ raw WaveSurfer recording
|
▼
Manual USV registration
|
▼
.jld
```

Previously generated `.jld` annotation files can also be reopened and edited.

3. Portable Repository Structure

Dependency files are now loaded relative to the location of `temp7.3.jl`.

Example:

```
include(joinpath(@__DIR__, "Necessities", "acoustic.jl"))
include(joinpath(@__DIR__, "Necessities", "gui_function_temp.jl"))
include(joinpath(@__DIR__, "Necessities", "GUItype.jl"))
include(joinpath(@__DIR__, "Necessities", "GDK_KEYmap.jl"))
include(joinpath(@__DIR__, "Necessities", "wavesurfer.jl"))
```

This replaces machine-specific absolute paths and makes the repository easier to clone and run on another computer.

Recommended directory structure:

```

USV_registration_GUI/
|
├─ temp7.3.jl
├─ usv.glade
|
└─ Necessities/
    ├─ acoustic.jl
    ├─ GUItype.jl
    ├─ gui_function_temp.jl
    ├─ GDK_KEYmap.jl
    ├─ wavesurfer.jl
    └─ JPEGsaver_function_temp.jl

```

4. JPEG Export Update

Saving annotations creates a `.jld` file and automatically exports individual USV spectrogram images.

Output directory:

```
<recording_name>_jpegs/
```

Default output size:

```
128 × 128 pixels
```

The current export pipeline uses:

- Quantile-based contrast normalization
- Low-frequency cutoff preprocessing
- Fixed 128 × 128 output
- Direct resizing for downstream image analysis

Example:

```

recording.h5
recording.jld
recording_jpegs/
├─ 0001_....jpeg
├─ 0002_....jpeg
├─ 0003_....jpeg
└─ ...

```

These images can be used directly for downstream USV image analysis and machine-learning pipelines.

5. GUI State Management

GUI state handling has been reorganized.

The GUI now separately maintains:

```
File state
├─ current filename
└─ saved / modified state

Data state
├─ spectrogram
├─ USV boxes
├─ time axis
└─ selected USV

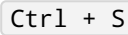
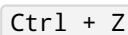
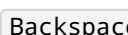
Interaction state
├─ drawing
├─ moving
└─ modifying
```

Drawing routines are primarily responsible for displaying the current state rather than determining it.

This reduces unwanted coupling between user interaction and screen rendering.

6. Keyboard and Mouse Controls

The keyboard controls introduced in previous versions remain available.

Input	Function
 / 	Move through recording
 / 	Select previous/next USV
Mouse wheel	Change displayed time scale
	Save annotation
	Delete selected USV
 / 	Delete selected USV
Left drag	Draw a new USV box
Modified left drag	Move a USV box
Right drag	Resize a USV box

The currently selected USV is highlighted separately from the remaining annotations.

7. Output

Saving generates:

```
recording.jld
```

containing serialized USV bounding-box information.

The GUI also calculates and displays USV features including:

- Duration
- Loudness
- Spectral purity
- Mean frequency
- Pitch variance

Population distributions of these features are displayed as histograms.

Summary

Temp7.2 → Temp7.3

Component	Temp7.2	Temp7.3
Filename in title bar	—	Added
Unsaved-change indicator	—	Added
Load another recording	Supported	Improved state reset
Dependency paths	Machine-specific	Repository-relative
Save-state tracking	—	Added
JPEG export	Full-band pipeline	Updated low-cut pipeline
GUI state organization	Mixed	More explicitly separated
Keyboard navigation	Supported	Retained
USV feature display	Supported	Retained

Temp7.3 primarily focuses on **workflow reliability, portability, and GUI state handling**, while preserving the manual USV registration functionality of Temp7.2.